RAKTABEEJ

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DATASET GENERATION ENGINE

"One Seed. Countless Descendants."

Developer: Arun VK

Browser-Native Dataset Generation Platform

Privacy-First AI Infrastructure

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WHY "RAKTABEEJ"?

From the **Devi Mahatmya** — the demon whose every drop of blood spawned another warrior.

THE MYTHOLOGICAL PARALLEL

The Legend: Raktabeej — a demon blessed with a terrible boon. Every drop of his blood that touched the earth birthed a new warrior, identical in power and ferocity.

The Parallel: One video uploaded to the platform generates thousands of dataset images — each frame a descendant of the original source.

The Philosophy: Multiplication of knowledge from a single source. One seed. Countless descendants.

1

↓

1,000+

One video spawns thousands of dataset descendants — each frame knowing its origin, every image traceable to the source.

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THE MODERN DATASET BOTTLENECK

Manual Frame Extraction

Tedious, error-prone, time-consuming manual work that drains engineering productivity.

Complex FFmpeg Commands

Steep learning curve requiring CLI expertise and command-line proficiency.

Python Scripting Required

Engineering team dependency for even simple dataset preparation tasks.

Privacy & Cost Barriers

Cloud services expose sensitive data. Enterprise tools carry prohibitive licensing costs.

Dataset generation should not require
a software engineering team.

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BROWSER-NATIVE GENERATION

VIDEO → SEGMENTATION → FRAME EXTRACTION → DATASET ARCHIVE → ML READY

THE THREE ZEROS

Zero Installation
Open the browser and begin

Zero Backend
No servers, no cloud dependency

Zero Data Leakage
Your data never leaves the device

CORE FEATURES

- Video Upload — Drag and drop any supported video format
- Frame Extraction — Precise interval-based capture
- Segment Processing — Time-based batch organization
- Dataset Packaging — Automatic ZIP archive generation
- Manifest Generation — Structured JSON metadata
- Browser-Only — Everything runs client-side

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ENGINEERING THE BLOODLINE

Technology Stack — Modern Engineering Meets Timeless Design

Layer	Technology	Purpose
UI Framework	React	Component-based user interface
Language	TypeScript	Type-safe development
Build Tool	Vite	Fast compilation and HMR
Styling	Tailwind CSS	Utility-first CSS framework
Video Processing	HTML5 Video API	Native browser video decoding
Frame Capture	Canvas API	Pixel-perfect frame extraction
Packaging	JSZip	Client-side archive generation

BROWSER-NATIVE

Entire Pipeline Runs Client-Side

Every step of the extraction pipeline executes within the browser environment:

- No server communication
- No cloud processing
- No data transmission
- No external dependencies

A self-contained engine that transforms video into ML-ready datasets entirely on the client.

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FROM SOURCE TO DESCENDANTS

01

Duration Detection

Native browser APIs parse video metadata and duration.

02

Segment Construction

Intelligent time-based segmentation of the video timeline.

03

Frame Extraction

Canvas API captures frames at precise, configurable intervals.

04

JPEG Encoding

Quality 0.85 encoding, optimized for ML training data.

05

Dataset Packaging

JSZip compresses frames into structured ZIP archives.

06

Manifest Generation

JSON metadata provides full traceability for every frame.

TECHNICAL SPECIFICATIONS

- **Native Browser APIs** — HTML5 Video API for decoding, Canvas API for frame capture, Blob API for binary handling
- **JPEG Quality 0.85** — Optimal balance between file size and image fidelity for machine learning training data
- **Real-Time Extraction** — Progressive frame capture with live progress tracking and memory-efficient streaming

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THE LIFECYCLE OF A DATASET

IDLE
Waiting for video upload

→

LOADING
Parsing video metadata

→

EXTRACTING
Frame capture in progress

→

DONE
Dataset ready for download

ERROR
Graceful recovery and cleanup

Controlled State Transitions: The system moves predictably through each phase. No state can be skipped, and every transition is validated. The **Error** state provides a safety net — if anything goes wrong at any stage, the system gracefully recovers, cleans up resources, and returns to Idle.

Abort support is available at any processing stage, ensuring the user always maintains control.

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WEAPONS OF THE ENGINE

INPUT	PROCESSING	OUTPUT
Video Upload Drag and drop video files directly into the browser interface.	**Live Progress** Real-time progress bar with segment and frame counters.	**Dataset ZIP** Complete packaged dataset with all extracted frames and metadata.
ZIP Support Import and process archived video collections in batch.	**Real-Time Gallery** Preview extracted frames as they are generated in the browser.	**Segment ZIP** Individual segment archives for modular dataset organization.
Multiple Formats MP4, WebM, MOV, and other popular video formats supported.	**Abort Support** Cancel extraction at any time with graceful resource cleanup.	**Manifest JSON** Structured metadata with full frame traceability and indexing.

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THE LINEAGE MANIFEST

FILENAME ANATOMY

video_segment_01_frame_001.jpg

- **Source Identifier** — The original video filename
- **Segment Index** — Which time segment the frame belongs to
- **Frame Index** — Sequential numbering within the segment
- **Extension** — JPEG format at quality 0.85

TRACEABILITY PRINCIPLES

Every frame in the dataset carries its complete ancestry. From any output image, you can trace back to the exact source video, the specific time segment, and the precise frame position.

This lineage ensures reproducibility, auditability, and scientific rigor in dataset management.

Each segment contains a sequence of frames, each with a unique index within its parent.

Every frame knows its origin.

WHERE IT CREATES VALUE

Computer Vision

Training data generation for image classification and recognition models.

Object Detection

Bounding box datasets from video frames for detection algorithms.

Search & Rescue

Frame extraction from drone and aerial surveillance footage.

Medical Imaging

Dataset preparation from scan and diagnostic video sequences.

Remote Sensing

Satellite imagery dataset generation for environmental and geological analysis.

Action Recognition

Temporal frame sequences for human activity and motion classification models.

Research

Academic dataset preparation without infrastructure investment or cloud dependency.

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RAKTABEEJ RESURRECTION

FROM DATASET ENGINE → TO DATASET INTELLIGENCE PLATFORM

STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5	STAGE 6
Frame Extraction	Dataset Manifest	AI Captioning	Object Detection	Auto Annotation	Training-Ready Export
✓	✓	IN DEV	PLANNED	PLANNED	VISION

RAKTABEEJ

ONE SEED.
COUNTLESS DESCENDANTS.

"Just as Raktabeej multiplied from a single drop, this platform transforms a single source into countless opportunities for AI training and discovery."

• FULLY DEPLOYED • BROWSER NATIVE • PRIVACY FIRST • ML READY • SCALABLE

Developer: Arun VK

Dataset Generation Engine

Raktabeej Resurrection Initiative